8	**(a)**		$H_0 : p = 0.81$ and $H_1 : p < 0.81$ p is the probability that a young adult (selected at random in England) has never donated blood	**B1** **B1** **[2]**	**3.3** **2.5**	H_0 : *probability* = 0.81 H_1 : *probability* < 0.81 Or proportion NOT number or how many young adults has never donated blood	
8	**(b)**		0.05	**B1** **[1]**	**1.2**	Accept 5%, 1/20 oe	

Question			Answer	Marks	AO	Guidance	
8	**(c)**		324	**B1** **[1]**	**1.1**	400×0.81 =324	
8	**(d)**		0.026 – 0.026125 **BC**	**B1** **[1]**	**1.1**	Percentages acceptable	
8	**(e)**		P($X \leq 310$) = 0.0448 – 0.045 [< 0.05] P($X \leq 311$) = 0.0576 – 0.058 [> 0.05] hence CR is [0 ≤] $x \leq 310$	***M1** ***M1** **DA1** **[3]**	**2.1** **1.1** **2.2a**	Additional calculations not penalised Accept x<311	**SC** If no marks scored and state $x \leq 310$ or x<311 B1,
8	**(f)**		314 is not in critical region accept H_0 there is insufficient evidence at the 5% level to **suggest** that the percentage of young adults (in England who have never given blood) is less than 81% **oe**	**M1** **A1** **A1** **[3]**	**1.1** **2.2a** **2.2b**	FT Comparison of 314 with *their* CR FT consistent with M mark allow "not significant" / "reject H_1 Not a **definite** statement, not prove Oe.Not enough evidence to suggest that the campaign has made more young people donate blood.	**or** P($X \leq 314$) = 0.114 > 0.05 SC If prob incorrect but correct comparison and consistent conclusion M1A0A0 If incorrect CR then could get M1A1A0